Blood Flow in the Microcirculation

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erythrocyte, red blood cell, capillary, hematocrit, suspension, cell-free layer

Abstract

The microcirculation is an extensive network of microvessels that distributes blood flow throughout living tissues. Reynolds numbers are much less than 1, and the equations of Stokes flow apply. Blood is a suspension of cells with dimensions comparable to microvessel diameters. Highly deformable red blood cells, which transport oxygen, have a volume concentration (hematocrit) of 40–45% in humans. In the narrowest capillaries, these cells move in single file with a surrounding lubricating layer of plasma. In larger vessels, the red blood cells migrate toward the centerline, reducing the resistance to blood flow. Vessel walls are coated with a layer of macromolecules that restricts flow. At diverging bifurcations, hematocrit is not evenly distributed in the downstream vessels. Other particles are driven toward the walls by interactions with red blood cells. These physiologically important phenomena are discussed here from a fluid mechanical perspective.

1. INTRODUCTION

RBC: red blood cell

Hematocrit: the volume fraction of red blood cells

ESL: endothelial surface layer

The circulatory system is a complex system of branching tubes, spanning a wide range of geometrical scales and flow velocities. Consequently, the mechanics of blood flow in the circulation encompasses a wide range of fluid mechanical phenomena. In large blood vessels, with diameters in the range of millimeters to centimeters, inertial effects are important, and phenomena such as boundary layers, flow separation, instability, secondary flows, and sometimes turbulence may be present. However, the fact that blood behaves essentially as a homogeneous Newtonian fluid in large blood vessels is a simplifying feature. The situation is reversed in the microcirculation. The Reynolds numbers (Re) of blood flow are typically much less than 1, so that inertial effects are negligible. For example, in a capillary with a diameter of 5 μm and flow velocity of 1 mm/s, Re is approximately 1.7×10^{-3}, based on a viscosity of approximately 0.03 dyn s/cm^2 and a density of approximately 1 g/cm^3. At the upper range of microvessel diameters, in an arteriole with a diameter of 100 μm and flow velocity of 2 cm/s, Re is approximately 0.7. Therefore, the fluid flow is described to a good approximation by the Stokes equations for an incompressible fluid. The linearity of these equations is an important simplifying feature. However, blood flowing in microvessels cannot be considered as a homogeneous fluid because it contains a high concentration of suspended cells, principally highly deformable red blood cells (RBCs; erythrocytes), whose dimensions are not much smaller than the blood vessel diameters. Noncontinuum effects become highly significant and must be considered. The study of the fluid mechanics of the microcirculation is largely concerned with examining the motion of a concentrated suspension of highly deformable particles in a geometrically complex branching network of very small tubes.

Figure 1 schematically illustrates several physiologically relevant phenomena associated with the motion of RBCs. In microvessels with diameters of 10 μm or more with normal hematocrits, multiple RBCs are present within each cross section of the vessel. The RBCs show a tendency to migrate away from the vessel walls, leading to the formation of a cell-free layer. This has the effect of reducing the resistance to flow, relative to what would be expected based on the bulk viscosity of blood. At diverging bifurcations, the hematocrit is distributed unevenly, with a higher hematocrit normally entering the branch with higher flow. This leads to a high degree of heterogeneity in individual vessel hematocrits within a network of microvessels and in some cases to the occurrence of plasma channels (i.e., capillaries without RBCs). In narrow capillaries, RBCs are highly deformed so as to squeeze through the vessels and often flow in single file. A further relevant phenomenon, not illustrated in the figure, is the presence of a glycocalyx or endothelial surface layer (ESL), consisting of a matrix of macromolecules that lines the microvessel walls and significantly restricts the flow in the vessels.

2. MECHANICAL PROPERTIES OF RED BLOOD CELLS

RBCs are specialized cells whose primary function is the transport of oxygen. When mammalian RBCs are formed, the nucleus is shed before the cell enters the bloodstream. In the absence of external stress, a normal human RBC has the shape of a biconcave disk with a diameter of approximately 8 μm and a thickness of approximately 2 μm. The interiors of RBCs contain a concentrated solution of hemoglobin, an oxygen-binding protein. When blood passes through the lungs, hemoglobin binds oxygen at a much higher concentration than would be reached in the absence of RBCs. Then, as the blood passes through the systemic circulation, oxygen is released to the surrounding tissues. The hemoglobin solution behaves as an incompressible fluid with a viscosity of approximately 7 cP (Chien 1975). In comparison, the viscosity of the suspending medium (plasma) is normally approximately 1.2 cP.

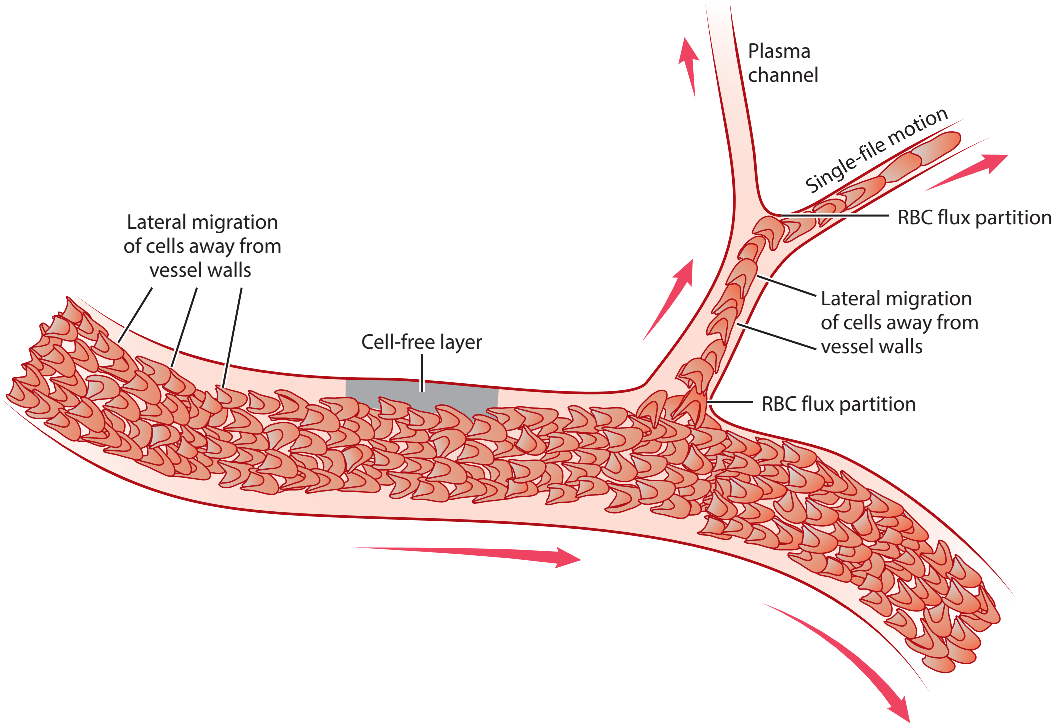

Figure 1

Fluid mechanical phenomena associated with red blood cells flowing in the microcirculation. Cells migrate laterally away from vessel walls, leading to a cell-free or cell-depleted layer. In diverging bifurcations, the partition of red blood cell flux is not necessarily in proportion to the partition of flow. Cells preferentially enter the high-flow branch, giving an increased hematocrit in that branch and a lower hematocrit in the other branch. There is single-file motion of red blood cells in narrow capillaries, and a plasma channel forms in a low-flow branch when all red blood cells enter the other branch.

From a mechanical perspective, the RBC membrane can be considered as a thin, highly deformable viscoelastic shell. It consists of a lipid bilayer and a mesh-like protein cytoskeleton lying on the inner surface of the bilayer. The tail regions of the lipid molecules are hydrophobic, and the lipid bilayer strongly resists area changes, which would result in the exposure of the lipid tails to the aqueous environment. The membrane therefore behaves as an almost incompressible material in two dimensions, with an elastic modulus of isotropic dilation of approximately 450 dyn/cm (Hochmuth & Waugh 1987). The mechanical response of the membrane to shear deformation is dominated by the protein cytoskeleton, which has viscoelastic properties (Evans & Hochmuth 1976). Estimates of the shear elastic modulus vary over a considerable range (Hochmuth & Waugh 1987). Recent work (Dimitrakopoulos 2012) suggests a value of approximately 2.5×10^{-3} dyn/cm, a lower value than was assumed in most previous studies. The shear viscosity has been estimated as 10^{-3} dyn s/cm (Hochmuth & Waugh 1987). The viscoelastic behavior of the membrane in shear can be represented by a Kelvin solid model, in which the total shear stress is the sum of viscous and elastic contributions (Evans & Hochmuth 1976). As a result of the fluid nature of the lipid bilayer, the cytoskeleton can move relative to the bilayer, as occurs when part of the membrane is stretched into a narrow tongue (Discher et al. 1994). The RBC membrane has a thickness less than 10 nm (Hochmuth et al. 1983), and the bending modulus is consequently very small, estimated as 1.8×10^{-12} dyn cm (Evans 1983).

In principle, this description of the mechanical properties of RBCs provides a sufficient basis for detailed simulations of the behavior of RBC suspensions under specified flow conditions. In practice, such simulations are challenging because of the complex mechanics of fluid-solid interactions, particularly with a large number of interacting particles. Examples of studies addressing this challenge are described in this review.

3. FLOW RESISTANCE AND APPARENT VISCOSITY

A quantity of obvious relevance when considering blood flow in tubes is the flow resistance R, defined as the pressure drop Δp along a tube divided by the volume flow rate Q. For steady laminar flow of a Newtonian fluid with viscosity μ, neglecting entrance effects, Poiseuille's law (Poiseuille 1846, Hagenbach 1860)

$$Q = \frac{\pi}{128}\frac{\Delta p D^4}{L\mu} \tag{1}$$

implies that

$$R = \frac{\Delta p}{Q} = \frac{128 L\mu}{\pi D^4}, \tag{2}$$

where L is the length and D the diameter. Flow resistance is proportional to the tube length and sensitively dependent on the diameter, being proportional to the inverse fourth power. This result is physiologically important, as it shows how relatively small changes in the diameter of microvessels can cause substantial changes in flow resistance and thereby control the distribution of blood flow to tissues at the local level. Such diameter changes can occur over short timescales (seconds to minutes) by active contraction and dilation of vascular smooth muscle surrounding microvessels, particularly arterioles, the smallest branches of the arteries (Secomb 2008). Over longer timescales (hours to days), vessel diameters are modulated by structural remodeling of the vessel walls, as occurs during growth, in response to changing metabolic needs, and in a number of disease states (Pries & Secomb 2014).

For the analysis of rheological effects associated with RBCs in microvessels, it is convenient to rearrange Equation 1 and define the apparent viscosity of blood by

$$\mu_{\mathrm{app}} = \frac{\pi}{128}\frac{\Delta p D^4}{LQ}. \tag{3}$$

The relative apparent viscosity is defined as $\mu_{\mathrm{rel}} = \mu_{\mathrm{app}}/\mu_{\mathrm{p}}$, where μ_{p} is the viscosity of the plasma or other suspending medium.

4. EXPERIMENTAL OBSERVATIONS OF BLOOD FLOW IN MICROVESSELS

Visual observation of the microcirculation became possible with the development of the microscope. Malpighi (1661) described the motion of blood through capillaries of the frog lung, thereby identifying the pathway for the circulation of the blood postulated by Harvey [1958 (1628)].

As mentioned above, the apparent viscosity of blood in microvessels may differ from the bulk viscosity owing to noncontinuum effects. Fåhraeus and Lindqvist passed human blood through narrow glass tubes of varying diameters and showed that the apparent viscosity decreases markedly with decreasing tube diameters below approximately 300 μm, a phenomenon known as the Fåhraeus-Lindqvist effect (Martini et al. 1930, Fahraeus & Lindqvist 1931, Pries et al. 1992, Secomb & Pries 2013). The tendency of RBCs to migrate away from vessel walls is largely responsible for this effect (Goldsmith 1971, Goldsmith et al. 1989). The trend of decreasing apparent viscosity continues as the diameter is decreased to approximately 6 μm, at which point the relative apparent viscosity is approximately 1.2 at normal hematocrit, less than half of its bulk value of approximately 3.0 (Pries et al. 1992). At still smaller diameters, the relative apparent viscosity rises again, as the tube diameter approaches the minimum that permits the passage of intact RBCs, and the layer of fluid around the cell becomes very narrow.

Discharge hematocrit: volume flux of red blood cells divided by total blood volume flux along a tube

Tube hematocrit: volume of red blood cells divided by total blood volume in a tube

Pries et al. (1992) collected and analyzed multiple experimental results on the flow of RBC suspensions in glass tubes and developed an empirical equation for the dependence of the relative apparent viscosity on the tube diameter and hematocrit:

$$\mu_{\mathrm{rel}} = 1 + (\mu_{45} - 1)\frac{(1 - H_{\mathrm{D}})^C - 1}{(1 - 0.45)^C - 1}, \tag{4}$$

where

$$\mu_{45} = 220\exp(-1.3D) + 3.2 - 2.44\exp(-0.06D^{0.645}) \tag{5}$$

is the relative apparent blood viscosity for a discharge hematocrit $H_{\mathrm{D}} = 0.45$, and D is the lumen diameter in micrometers. The coefficient C giving the dependence on hematocrit is governed by

$$C = (0.8 + e^{-0.075D})\left(-1 + \frac{1}{1 + 10^{-11}D^{12}}\right) + \frac{1}{1 + 10^{-11}D^{12}}. \tag{6}$$

A further consequence of the tendency of RBCs to migrate away from vessel walls is a phenomenon known as the Fåhraeus effect (Fahraeus 1928), in which the tube hematocrit, H_{T}, is less than the discharge hematocrit, H_{D}. Being concentrated toward the center of the tube, the RBCs travel faster on average than the surrounding plasma and therefore have shorter transit times. The reduction in hematocrit is related to the mean velocity V_{RBC} of RBCs within the tube and the overall mean flow velocity V_{bulk} according to the equation (Sutera et al. 1970)

$$\frac{H_{\mathrm{T}}}{H_{\mathrm{D}}} = \frac{V_{\mathrm{bulk}}}{V_{\mathrm{RBC}}}. \tag{7}$$

An empirical relationship was obtained for the dependence of the Fåhraeus effect on the tube diameter and discharge hematocrit (Pries et al. 1992):

$$\frac{H_{\mathrm{T}}}{H_{\mathrm{D}}} = H_{\mathrm{D}} + (1 - H_{\mathrm{D}}) \cdot (1 + 1.7 \cdot e^{-0.35D} - 0.6 \cdot e^{-0.01D}). \tag{8}$$

For very narrow tubes, this ratio is close to 1.0. It drops to a minimum value at a diameter of approximately 13 μm. For a discharge hematocrit of 0.45, the minimum value is approximately 0.72. The ratio increases toward 1.0 at larger diameters.

Methods for quantitative measurements of blood flow in microvessels in living tissues advanced in the twentieth century, including those for determining the flow velocity of RBCs (Intaglietta et al. 1970) and pressure in microvessels (Wiederhielm et al. 1964). In principle, these methods would allow the estimation of flow resistance in microvessels in vivo. In practice, such measurements proved to be technically very difficult because of the need to measure the small pressure difference between two points along a single microvessel. The few such data available suggested that the apparent viscosity in vivo was higher than expected based on glass tube experiments (Lipowsky et al. 1978, 1980).

Pries et al. (1990, 1994) took an indirect approach in which they examined the distributions of blood flow and hematocrit in microvascular networks of the rat mesentery containing several hundred segments. These data were compared with the predictions derived from a theoretical model of flow in networks. An empirical relationship to describe the dependence of the apparent viscosity on the vessel diameter and hematocrit was derived as follows:

$$\mu_{\mathrm{rel}} = \left[1 + (\mu_{45} - 1)\frac{(1 - H_{\mathrm{D}})^{C} - 1}{(1 - 0.45)^{C} - 1}\left(\frac{D}{D - 1.1}\right)^{2}\right]\left(\frac{D}{D - 1.1}\right)^{2}, \tag{9}$$

where

$$\mu_{45} = 6\exp(-0.085D) + 3.2 - 2.44\exp(-0.06D^{0.645}) \tag{10}$$

is the relative apparent blood viscosity for $H_{\mathrm{D}} = 0.45$ and the quantity C is given by Equation 6. According to this result, the apparent viscosity in living microvessels with a diameter of 30 μm or less is much higher than its value in glass tubes with corresponding diameters. The main cause of this difference was found to be the presence of an ESL, or glycocalyx, on the inner surface of endothelial cells, with a width of the order of 1 μm (Pries et al. 2000). This layer consists of a dilute matrix of loosely bound macromolecules, which largely impedes plasma flow in its interior and thereby reduces the effective width of the lumen available for the flow of plasma and RBCs.

During the past 20 years, the above equations for the apparent viscosity of blood in vitro and in vivo have been used extensively in theoretical analyses of blood flow in networks of microvessels. However, these equations are empirical and not derived from analyses of the fluid mechanical phenomena involved. The development, from first principles based on knowledge of the mechanical properties of RBCs and other components of the system, of theories capable of predicting the behaviors described by these equations has proved to be a formidable challenge. Progress toward this goal is discussed next. For a more detailed review of techniques for the numerical simulation of RBC motion in suspension flows, readers are referred to Freund (2014).

5. ANALYSIS OF RED BLOOD CELL MOTION IN CAPILLARIES

When RBCs flow through capillaries with diameters less than approximately 6 μm, they normally move in a single file. For capillaries with diameters in the range 6–10 μm, a hematocrit-dependent transition from single-file to multifile flow is observed, with single-file flow at low hematocrits (Gaehtgens et al. 1980). A number of simplifying assumptions can be made in analyzing the fluid mechanics of single-file RBC motion in capillaries (Secomb et al. 1986). First, for Stokes flow in a narrow tube, the perturbation to the parabolic velocity profile resulting from a suspended particle decays rapidly with distance from the particle along the tube. It is therefore a reasonable approximation to consider the motion of a single RBC, neglecting cell-cell interactions. Second, an RBC in a narrow capillary almost fills the tube, so lubrication theory can be used to describe the fluid flow in the narrow gaps between the RBC and the wall. Third, the deformed shapes that RBCs assume in narrow capillaries have approximate rotational symmetry about the tube axis, and the assumption of axisymmetric shapes can be justified.

Early analyses of single-file RBC motion in capillaries date back to the late 1960s (Barnard et al. 1968, Lighthill 1968, Fitz-Gerald 1969). In the further development of this approach, the mechanical properties of RBCs were represented in increasingly realistic fashion (Lin et al. 1973, Zarda et al. 1977). The application of finite element methods allowed the analysis of the fluid flow without assuming lubrication theory (Zarda et al. 1977).

If the cell shape is axisymmetric, then it quickly approaches a steady shape with no internal motion when flowing in single-file flow in a narrow uniform tube. The membrane can therefore be considered as a purely elastic shell, even though it is viscoelastic. The stresses in the membrane can then be expressed as functions of the cell deformation, based on its mechanical properties, as discussed above (Evans & Skalak 1980). Because the membrane deforms at almost constant area, the isotropic component of in-plane membrane stress is treated as a free variable, analogous to

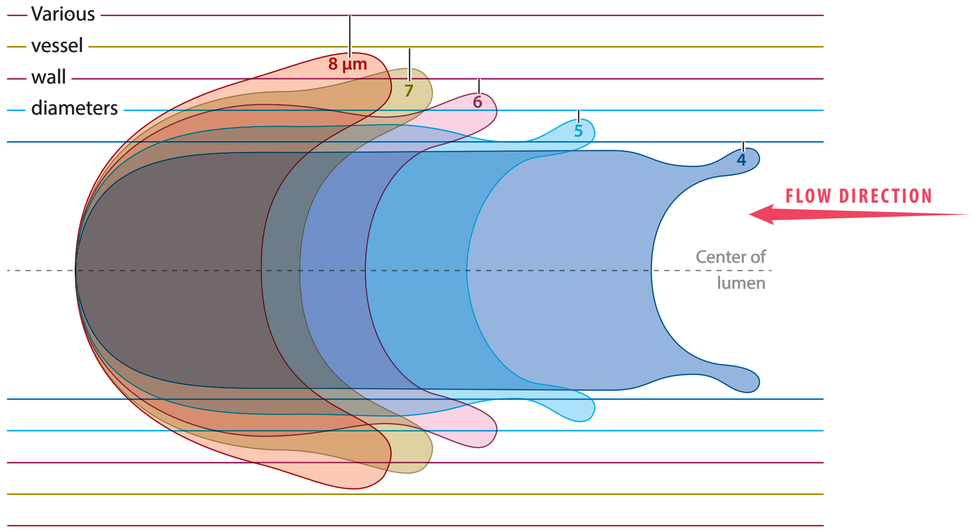

Figure 2

Computed axisymmetric red blood cell shapes in vessels with diameters ranging from 4 μm to 8 μm. The corresponding three-dimensional cell shapes are obtained by rotating around the central axis. The cell velocity is 0.01 cm/s. Figure adapted with permission from Secomb (1987).

the pressure in an incompressible fluid. The equations of mechanical equilibrium are those of a thin axisymmetric shell (Timoshenko 1940). The external fluid stresses acting on the shell can be calculated using lubrication theory (Cameron 1966) as functions of the cell shape and velocity. Combining the constitutive equation of membrane elasticity, the equilibrium equations for an axisymmetric shell and lubrication equations yield a system of nonlinear ordinary differential equations (Secomb et al. 1986). In these equations, the variables are the coordinates of a point on the membrane, the angle of the membrane to the axis, the curvature, transverse shear stress and tension components in the membrane, and the pressure difference across the membrane, all expressed as functions of arc length measured from the point where the front of the cell meets the tube axis. This system can be solved numerically as a boundary value problem.

Figure 2 shows results of this analysis (Secomb 1987). The predicted RBC shapes show several features that agree with experimental observations (Skalak & Branemark 1969, Gaehtgens et al. 1980). At the front (leading portion) of the cell, the shape is convex and smoothly rounded, whereas at the rear it is concave with rapid changes in curvature. The rim of the concave region typically shows a small outward bulge toward the rear of the cell.

According to the assumptions of this analysis, the increase in the relative apparent viscosity due to the presence of RBCs is proportional to the hematocrit H_T, and the constant of proportionality K_T (the apparent intrinsic viscosity) can be deduced from the pressure drop over the length of the cell for any given flow condition. The value of K_T increases with decreasing tube diameter, because the cell fills the tube more fully, and also decreases with decreasing velocity, because the cell shape becomes less streamlined and approaches the wall more closely. Good quantitative agreement was demonstrated between predicted K_T values and those deduced from experimental studies, as a function of tube diameter and cell velocity (Secomb et al. 1986, Secomb 1987).

In reality, RBC shapes in capillaries are not necessarily axisymmetric, even in single-file flow (Skalak & Branemark 1969, Tomaiuolo et al. 2009). The resulting asymmetry of fluid forces

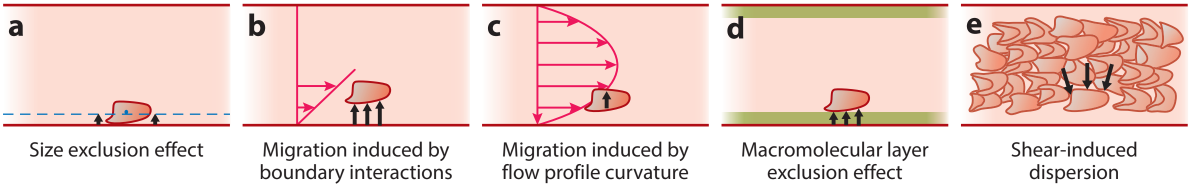

Figure 3

Mechanisms causing lateral migration of red blood cells (RBCs) flowing in microvessels. (*a*) The RBC center of mass (*blue dot*) is excluded from a zone near the wall (*blue dashed line*) as a result of the finite size of the RBC. (*b*) RBCs and other deformable particles tend to migrate away from solid boundaries when suspended in a shear flow. (*c*) The curvature of the velocity profile in a tube flow induces inward migration, independent of the influence of solid boundaries. (*d*) Living microvessels are lined with a macromolecular layer (*green*) that impedes plasma flow and largely excludes flowing RBCs. (*e*) In shear flow of concentrated suspensions, particle-particle interactions induce random lateral particle motions, whose net effect is to produce motion down the concentration gradient, toward the wall. This effect is known as shear-induced dispersion.

acting on the RBC leads to tank treading (Gaehtgens & Schmid-Schönbein 1982), a phenomenon seen when RBCs are suspended in shear flows (Fischer et al. 1978). Early theoretical analyses of this phenomenon used a two-dimensional model (Secomb & Skalak 1982) or a prescribed three-dimensional cell shape (Hsu & Secomb 1989). More recent computational techniques have permitted fully three-dimensional simulations for deformable RBCs (Noguchi & Gompper 2005; Doddi & Bagchi 2009; Fedosov et al. 2010a,b; Freund & Orescanin 2011), including predictions of flow resistance, tank-treading frequency, and lateral offset of an asymmetric RBC from the tube axis in capillary flow (Freund 2014).

6. RADIAL DISTRIBUTION OF RED BLOOD CELLS IN MULTIFILE FLOW

The formation of a cell-free or cell-depleted layer near the vessel walls is the main reason for the distinctive flow properties of blood in microvessels. Several physical phenomena contribute to this behavior (**Figure 3**). For example, the finite size of the RBC limits the radial distribution of cell centers (**Figure 3*a***). For an RBC in a disk-like configuration, the minimum dimension is at least 2 μm, so the center of mass of the cell cannot physically approach within 1 μm of the wall. When placed in a shear flow adjacent to a solid boundary, RBCs show a tendency to migrate away from the boundary (**Figure 3*b***). Independent of wall effects, the curvature of the Poiseuille velocity profile in tube flow produces a tendency for migration toward the centerline of the flow (**Figure 3*c***). In microvessels in vivo, an additional effect comes into play that is not present in glass tubes. The glycocalyx or ESL, consisting of a matrix of macromolecules, creates an exclusion zone for flowing RBCs (**Figure 3*d***). In a shear flow of a concentrated suspension, the frequent particle-particle interactions lead to a net migration down the concentration gradient, toward the walls (**Figure 3*e***). This effect acts in opposition to the mechanisms driving migration away from the wall.

Tank treading: continuous motion of the membrane around the surface of a red blood cell or other deformable particle suspended in shear flow, with the particle maintaining a constant or almost constant shape

6.1. Migration Away from Solid Boundaries

For Stokes flow of suspensions in tubes, migration away from the walls is a consequence of particle deformability (Goldsmith 1971, Grandchamp et al. 2013). A neutrally buoyant rigid particle in dilute suspension does not generally migrate across the flow away from a solid boundary in Stokes flow. For spherical particles, this result follows from the fact that equations of fluid motion are linear and that a reversal of the flow direction would result in the opposite migration, which would be a contradiction. For nonspherical particles, the same argument applies if the particle and flow have symmetry with respect to a plane perpendicular to the flow direction at some point during motion (Secomb 2003). For instance, a rigid ellipsoidal or disk-shaped particle typically tumbles when placed in a shear flow, and thus satisfies this condition.

This migration is demonstrated in three-dimensional simulations of motions of RBCs and other deformable particles in shear flow near a solid boundary (Coupier et al. 2008, Doddi & Bagchi 2009). However, a general mechanistic understanding of the phenomenon has proved elusive. In particular, the direction of migration for deformable RBCs and vesicles close to the wall is apparently away from the wall in all reported experiments and simulations, but no proof has been presented that this is always true for such particles. Several theoretical studies have examined particular cases.

Olla (1997) analyzed the motion of an ellipsoidal particle with a specific assumed tank-treading membrane motion (Keller & Skalak 1982) and demonstrated a drift away from the wall. In this case, the particle assumes an orientation in which its long axis is inclined at a positive angle to the x axis (assuming flow in the x direction with a positive velocity gradient in the y direction). The interaction of the flow perturbation caused by the particle with a rigid boundary was shown to produce drift away from the wall. The estimate of the drift velocity was based on the image method for Stokes flow, which is applicable for particles far from the wall. For particles close to the wall, the conclusion that a positive inclination to the wall causes drift away from the wall can be obtained by lubrication arguments (Secomb & Hsu 1993).

For RBCs dilutely suspended in shear flow, stable tank-treading motions are observed only for high-viscosity suspending media. If the suspending medium has a viscosity similar to that of plasma, RBCs exhibit tumbling motions with the vorticity vector of the shear flow as the axis (Keller & Skalak 1982, Barthes-Biesel 2016). Migration away from the wall is still observed in this case (Grandchamp et al. 2013). Olla (1999, p. 454) speculated that "the particle undergoes a kind of rigid rotation which privileges orientations corresponding to drift away rather than towards the wall." Hariprasad & Secomb (2014) considered this phenomenon using a previously developed two-dimensional model for RBC motion in flow that incorporates effects of membrane viscoelasticity (Secomb et al. 2007). To investigate the mechanisms of lift generation, they computed cell trajectories under lateral forces directed toward the vessel wall. For a sufficiently large force, the cells assumed a stable configuration close to the wall with the major axis oriented at a small positive angle to the wall and tank-treading membrane motion. The mechanism of lift generation was then as described in the previous paragraph. At a reduced lateral load, the cells exhibited cyclic tumbling motions. In these motions, the major axis of the cells had a small positive angle to the wall during most of the cycle, with rapid excursions during the tumbles, as suggested by Olla (1999). These studies give some useful insights but provide only a partial answer to the challenge of understanding the mechanisms of lateral drift of cells and vesicles near boundaries in Stokes flow, and more work is needed in this area.

6.2. Migration Resulting from the Curvature of the Velocity Profile

The mechanism discussed in the previous section would occur in the case of a linear shear flow over a solid boundary. A distinct mechanism for migration arises from the fact that the fluid motion in tube flow has a curved velocity profile, with a velocity gradient that increases from the centerline to the wall. Kaoui et al. (2008) performed a numerical study of the migration of a vesicle in an unbounded Poiseuille flow and demonstrated migration toward the centerline. Further work using a two-dimensional model for RBCs with membrane bending (but not shear) elasticity showed that the cells may approach asymmetric shapes, with the equilibrium position of the center of mass displaced from the centerline (Kaoui et al. 2009). The occurrence of this behavior depends on the cell velocity and on its perimeter relative to its area. Shi et al. (2012) also used a two-dimensional model for RBCs, for stiffened cells at relatively high Reynolds numbers at which inertial effects are significant. The combined influence of wall effects and velocity profile curvature on particle migration was analyzed experimentally and theoretically for vesicles (Coupier et al. 2008). In experimental studies, it is difficult to separate wall effects from the effects of velocity profile curvature. The effect of velocity profile curvature likely plays a significant role in the migration of RBCs to the centerline in dilute suspensions. In blood or other concentrated suspensions, this effect is probably dominated by shear-induced dispersion (as discussed below) in the interior region of the flow and by effects of wall interactions in the peripheral part of the flow.

6.3. Effects of the Endothelial Surface Layer

The walls of living microvessels are lined with a layer of macromolecules attached to endothelial cells that form the inner surface of the vessel walls. This layer is referred to as the glycocalyx or ESL (Pries et al. 2000, Reitsma et al. 2007, Weinbaum et al. 2007). The thickness is variable but is estimated to be of order 1 μm. From a mechanical perspective, this dilute gel-like layer may be regarded as a deformable porous medium. The hydraulic resistance has been estimated to be at least 10^{11} N s/m^4 (Secomb et al. 1998), which is low relative to other biological materials but is sufficient to largely inhibit plasma flow within the layer. Although static RBCs can compress the layer, flowing RBCs are excluded by a lubrication-type mechanism, which has been likened to skiing on powder snow (Feng & Weinbaum 2000; Secomb et al. 2001a,b). Endothelial cells are sensitive to wall shear stress, and the presence of the ESL has the significant implication that the shear stress acting on the endothelial cell membrane is transmitted mainly by the ESL, not by fluid adjacent to the membrane (Secomb et al. 2001a,b). With regard to blood flow in microvessels, the dominant effect of the ESL is to reduce the effective diameter available for flow, and therefore to substantially increase the flow resistance, as discussed above. Although the mechanics of interactions between suspended particles and a deformable porous wall layer is an interesting and biologically relevant topic, it has received relatively little recent attention (Beaucourt et al. 2004, Hariprasad & Secomb 2012).

6.4. Migration due to Cell-to-Cell Interactions

The mechanisms just discussed lead to the formation of a cell-free or cell-depleted layer at the wall. In dilute suspension, RBCs are eventually driven to the centerline. Under physiological conditions with a hematocrit of 40–45%, however, this trend is obviously opposed by the effect of RBCs crowding in the interior of the vessel (Goldsmith 1971, Secomb 2003, Pranay et al. 2012). In a suspension subject to shear flow, the high concentration leads to frequent collisions (i.e., strong hydrodynamic interactions between particles), which cause fluctuations in lateral velocity and drive a diffusion-like motion of the particles, with a net flux down the concentration gradient. This effect is termed shear-induced diffusion or dispersion (Leighton & Acrivos 1987). The width of the cell-free or cell-depleted layer is governed by the balance between the various mechanisms driving the formation and enlargement of the layer and the opposing effects of shear-induced diffusion.

Presently, there is no comprehensive theory to predict shear-induced diffusion in concentrated suspensions of deformable particles. Scaling arguments (Leighton & Acrivos 1987, Pranay et al.

2012) imply that the net RBC flux driven by a gradient in concentration is governed by a gradient diffusivity

$$D_{\mathrm{g}} = \dot{\gamma}\varphi a^2 f_{\mathrm{s}}, \tag{11}$$

where $\dot{\gamma}$ is the shear rate, φ is the volume fraction, a is a typical particle dimension, and f_{s} is a dimensionless parameter that depends on particle shape and deformability. (This should be distinguished from self-diffusivity, which describes the growth with time of the mean squared displacement of an individual RBC.) The parameter f_{s} can be estimated from experimental observations of RBC suspensions (Grandchamp et al. 2013). In principle, a theory of this type for shear-induced diffusion, along with theories as discussed above for migration away from solid boundaries, would provide a basis for quantitative modeling of the dynamic properties of the cell-free layer as well as its eventual equilibrium width. Some progress has been made in this direction (Pranay et al. 2012, Hariprasad & Secomb 2014), but more work is needed. An alternative theoretical approach is the direct simulation of flow of multiple interacting particles in specific geometries, as described below.

6.5. Effect of Cell-Free Layer on Flow Resistance

The substantial reduction in flow resistance in a tube resulting from the presence of a cell-free or cell-depleted layer near the wall can be understood by considering that the rate of energy dissipation per unit volume in a viscous fluid is given by $\mu\dot{\gamma}^2$, where the shear rate $\dot{\gamma}$ is maximal near the tube wall and zero at the centerline. Any redistribution of RBCs leading to a reduction in viscosity near the wall can significantly reduce overall energy dissipation. A quantitative estimate of the effect can be obtained by using a simple two-phase model (Vand 1948) (**Figure 4**). In this model, a cylindrical core region of viscosity μ_{c} centered on the tube axis is surrounded by a cell-free layer of lower viscosity μ_{p}. The velocity profile is given by

$$v(r) = \begin{cases} \dfrac{\Delta p a^2}{4L}\left(\dfrac{\lambda^2 - r^2/a^2}{\mu_c} + \dfrac{1-\lambda^2}{\mu_{\mathrm{p}}}\right) \text{ when } r < \lambda a \\ \dfrac{\Delta p a^2}{4L}\dfrac{1 - r^2/a^2}{\mu_{\mathrm{p}}} \text{ when } r \geq \lambda a \end{cases}, \tag{12}$$

where $\lambda = 1 - \delta/a$ is the ratio of the core radius to the tube radius, δ is the width of the cell-free layer, and a is the tube radius. The apparent viscosity as defined in Equation 3 is given by

$$\mu_{\mathrm{app}} = \frac{\mu_{\mathrm{p}}}{1 - \lambda^4(1 - \mu_{\mathrm{p}}/\mu_{\mathrm{c}})}. \tag{13}$$

With a cell-free layer of only 10% of the vessel radius, one obtains $\lambda = 0.9$, $\mu_{\mathrm{c}} = 3\mu_{\mathrm{p}}$, and the apparent viscosity is $\mu_{\mathrm{app}} = 1.78\mu_{\mathrm{p}}$, which is substantially less than the viscosity in the core region. This formula gives results in close agreement with experimental data (Pries et al. 1992) for glass tubes with diameters in the range 30–1,000 μm, if it is assumed that the cell-free layer has a constant width of 1.8 μm, independent of the tube diameter (Secomb 1995). The independence of the layer thickness on the vessel diameter is to be expected if the layer width is determined by the local balance between inward and outward migration tendencies, as discussed above. The 1.8-μm width of the layer was, however, obtained by fitting the experimental results and not from theoretical arguments.

In recent years, advances in numerical techniques and in parallel computation capabilities have made it feasible to perform direct simulations of multiple RBCs flowing in narrow tubes, including a fairly realistic representation of the cells' mechanical properties (Dupin et al. 2007;

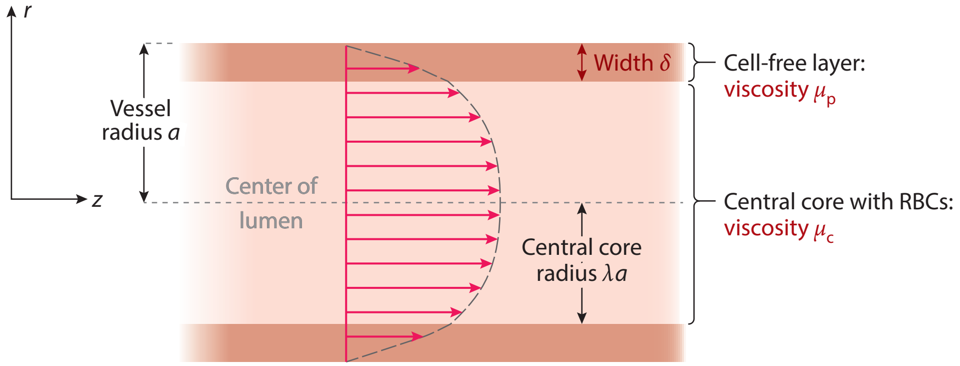

Figure 4

Two-phase model for blood flow in a microvessel. A typical resulting velocity profile is shown.

McWhirter et al. 2009; Fedosov et al. 2010a,b; Freund & Orescanin 2011; Freund 2014). From such simulations, the apparent viscosity of the flow can be deduced, and the statistical properties of the time-varying cell-free layer thickness can be analyzed. In simulations by Fedosov et al. (2010a,b), the predicted relative apparent viscosity agreed closely with experimental results (Pries et al. 1992) for hematocrits between 15% and 45% and tube diameters from 10 to 40 μm. Moreover, the average cell-free layer thickness agreed with the estimated value of 1.8 μm mentioned above (Katanov et al. 2015). These findings support the view that the flow properties of blood in narrow glass tubes can indeed be understood in terms of the mechanical properties of individual RBCs and principles of fluid mechanics. At the same time, this approach requires a separate, computationally intensive simulation for each case of interest, and fundamental fluid mechanical mechanisms may be difficult to deduce from the results. Significant challenges remain in developing a general understanding of blood flow in narrow tubes.

7. RED BLOOD CELL FLUX PARTITION IN DIVERGING BIFURCATIONS

The microcirculation has complex network geometry. RBCs pass through many diverging and converging bifurcations during each passage through the circulatory system. The presence of cell-free layers in microvessels strongly influences the distribution of RBCs between the two daughter branches of a diverging bifurcation. If one branch has a lower flow rate than the other, it draws its flow more from the periphery of the bloodstream in the parent vessel and therefore receives disproportionally fewer RBCs (**Figure 5*a***). The hematocrit is thus reduced in the low-flow branch and increased in the high-flow branch, relative to the hematocrit in the parent vessel.

Experimental observations of the hematocrit partition in microvessel bifurcations in the rat mesentery (Pries et al. 1989) were used to derive a set of empirical equations for the dependence of the RBC fluxes in the daughter branches on the flow fraction in each vessel, the vessel diameters, and the hematocrit in the parent vessel. These equations were later refined (Pries & Secomb 2005) to avoid unphysical predictions for extreme values of the hematocrit and diameter. The fraction FQ_E of RBCs entering one branch is given as a function of the fraction FQ_B of the total flow in that branch:

$$\operatorname{logit} FQ_E = A + B \operatorname{logit}[(FQ_B - X_0)/(1 - X_0)], \tag{14}$$

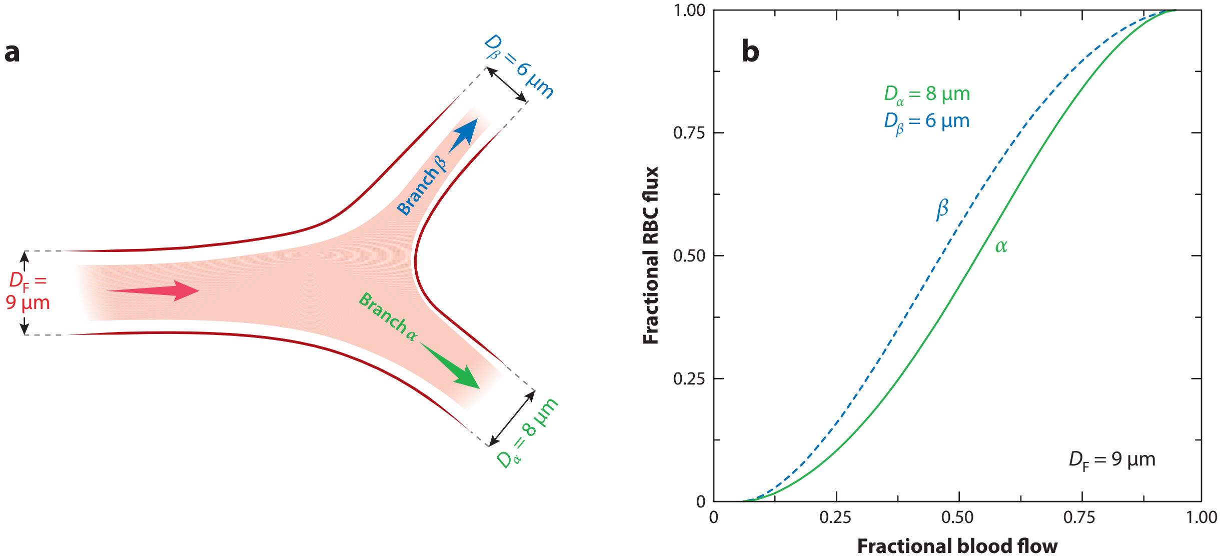

Figure 5

Red blood cell (RBC) partition in a diverging microvessel bifurcation, with vessel diameters as indicated. (*a*) Example of bifurcation geometry, showing a typical distribution of RBCs (*pink*) when the flow in branch β is a small fraction of the total flow. (*b*) RBC flux fraction in each branch as a function of the overall flow fraction entering that branch. Curves are derived from empirical relationships, assuming a discharge hematocrit of 0.4 in the parent vessel.

where logit $x = \ln(x/(1-x)]$ and

$$A = -13.29\left[\left(D_\alpha^2/D_\beta^2 - 1\right)/\left(D_\alpha^2/D_\beta^2 + 1\right)\right](1 - H_{\mathrm{D}})/D_{\mathrm{F}}, \tag{15}$$

$$B = 1 + 6.98(1 - H_{\mathrm{D}})/D_{\mathrm{F}}, \tag{16}$$

$$X_0 = 0.964(1 - H_{\mathrm{D}})/D_{\mathrm{F}}, \tag{17}$$

where D_α, D_β, and D_{F} are the diameters of the two branches and the parent vessel, and H_{D} is the discharge hematocrit in the parent vessel.

Figure 5 illustrates the resulting behavior for a parent vessel 9 µm in diameter feeding branches with diameters of 6 and 8 µm. The minimum flow fraction in either branch for RBCs to enter that branch is $X_0 = 0.064$. The RBC flux fraction in each branch increases nonlinearly with the flow fraction entering that branch. For a given flow fraction, the smaller-diameter branch receives a higher hematocrit. This behavior occurs because a small-diameter branch draws its flow from a small part of the periphery of the parent vessel, and more of its flow comes from the high-hematocrit core region of the parent vessel, compared to a large-diameter branch.

As this discussion indicates, the main aspects of RBC flux partition in diverging bifurcations can be understood, at least qualitatively, by considering the shape and RBC content of the regions in the upstream cross section of the parent vessel that supply each of the daughter vessels. In such considerations, it is implicitly assumed that the RBCs approximately follow the streamlines of the underlying flow of the suspending medium. This is not necessarily the case, particularly when the suspended particles are relatively large compared to the vessel diameters. In early studies of this topic (Fung 1973), it was considered that RBCs show an attraction (i.e., migration) toward the high-flow-rate branch (Doyeux et al. 2011). In a study using a two-dimensional model of deformable RBCs, Barber et al. (2008) identified two opposing tendencies of cells to migrate across streamlines of the underlying flow. On the one hand, the flow streamline that arises from the centerline of the parent vessel almost certainly enters the high-flow branch, so migration toward the centerline tends to increase the hematocrit in the high-flow branch. On the other hand, an obstruction effect was identified, in which RBCs that closely approach a low-flow branch can be drawn into that branch even when the streamline of the underlying flow corresponding to the cells' center of mass would enter the high-flow branch. Subsequently, Doyeux et al. (2011) used microfluidic experiments and two-dimensional simulations to demonstrate such an attraction toward the low-flow branch.

The theoretical simulations referenced above are for RBCs in dilute suspension. The RBC flux partition behavior in bifurcations depends on the hematocrit in the parent vessel. As the parent hematocrit approaches 100%, the disparity in hematocrits in daughter vessels decreases, given that both must approach 100%. This behavior is enforced in the empirical Equations 14–17. The effects of two-cell interactions on RBC flux partition were explored using two-dimensional simulations (Barber et al. 2011). The most common type of interaction was termed a trade-off, in which a cell entering one branch causes a following cell to enter the other branch. This tends to equalize hematocrit in the two branches. However, other types of interactions occurred for some initial cell positions, such as herding, in which the leading cell is caused to enter the same branch as the following cell, and following, in which the trailing cell is caused to enter the same branch as the leading cell. Even for two-cell interactions, the parameter space is large and systematic investigation is difficult, and these difficulties are magnified when multicell interactions are considered. Further progress in understanding the fluid mechanics of RBC flux partition will probably require a more phenomenological approach based on large-scale numerical simulations (Yin et al. 2013, Lykov et al. 2015).

8. INTERACTIONS OF RED BLOOD CELLS WITH OTHER SUSPENDED PARTICLES

Although RBCs are by volume the dominant suspended particles in blood, other types of cells and particles are also biologically important. White blood cells (leukocytes) of several types are essential to the body's responses to injury and infection. They are generally larger than RBCs and less deformable. Platelets (thrombocytes) are cellular fragments, with a typical maximum dimension of 2–3 μm, whose main function is to prevent bleeding from damaged blood vessels. Lipids are carried in the bloodstream in protein capsules, forming particles called chylomicrons. The size of these particles varies widely, with a maximum of approximately 0.6 μm. One method for administering drugs via the bloodstream involves the use of liposomes, artificial capsules formed from phospholipids. These can be made in a range of sizes, from tens of nanometers up to a few micrometers.

For all the particle types mentioned above, the ability to reach microvessel walls is essential for them to perform their biological functions. If the particle size is much less than 1 μm, Brownian motion ensures that the particles are well distributed throughout the vessel cross section. For larger particles, lateral motion within a microvessel depends on fluid dynamic interactions with other particles, primarily RBCs. Experimental observations of flow in narrow tubes show that both leukocytes and platelets are preferentially distributed near the vessel walls (Goldsmith & Spain 1984, Nobis et al. 1985, Tangelder et al. 1985). This margination of leukocytes and platelets is attributed to hydrodynamic interactions with RBCs. Such behavior can be explained by the fact that RBCs are much more deformable than these other types of particles. As discussed above, the migration of RBCs away from vessel walls is dependent on their deformability. The relatively strong inward migration of RBCs induces outward migration of more rigid particles (Kumar & Graham 2012). Several computational studies have confirmed this general concept and provided quantitative data on the margination of leukocytes (Freund 2007), platelets (Vahidkhah et al. 2014, Mehrabadi et al. 2015), and micro- and nanoparticle drug carriers (Muller et al. 2014).

9. CONCLUSIONS AND OUTLOOK

Blood flow in the microcirculation is characterized by Reynolds numbers much less than 1, such that inertial effects are negligible and the equations governing fluid flow are linear. However, complexity arises from the fact that blood is a concentrated suspension of RBCs, highly deformable particles with dimensions comparable to microvessel diameters. The flow-driven interactions of RBCs with each other, with flow boundaries, and with other suspended particles give rise to a number of physiologically important fluid dynamical phenomena, including the formation of a cell-depleted layer near the walls, reduction in apparent viscosity, unequal partition of hematocrit at diverging bifurcations, and margination of other more rigid particles.

The physical mechanisms underlying these phenomena are in general fairly well understood in qualitative terms. The mechanical properties of individual RBCs have been much studied, and the values of key parameters describing elastic and viscous properties have been estimated. In principle, this information provides a basis for quantitative predictions of blood flow behavior in microvascular flow geometries. For the simplest cases, such as the flow of a single RBC in a narrow capillary, this goal was substantially achieved decades ago. However, the simulation of multiple interacting deformable RBCs in three dimensions presents formidable challenges, and progress was limited until comparatively recently. During the past 10 years, this area has seen a remarkable level of activity and progress, as described in this review, making use of increasingly powerful computers and newer numerical techniques (Freund 2014).

An important challenge for the future is the development of quantitative theories that will allow reliable prediction of physiologically relevant flow phenomena in the microcirculation, without the need to simulate the motion of every individual RBC. Such theories will require a better understanding of the underlying fluid mechanical phenomena, particularly those governing the radial distribution of RBCs in microvessels. The availability of large-scale multicell simulations is likely to be valuable in the formulation and testing of such theories. Improved techniques for analyzing and interpreting the results of the simulations may be needed for this.

Most of this recent work using multicell numerical simulations has considered normal RBCs in geometries with rigid impermeable boundaries. Such boundary conditions are appropriate for microfluidic devices, but not for microvessels in vivo. An area meriting more study is the multifile motion of RBCs in microvessels and bifurcations lined with an ESL. The effects of altered RBC properties, as occur in sickle-cell anemia, malaria, and diabetes, should be further investigated. The results of fluid mechanical analyses should be combined with knowledge about biological interactions of suspended elements with the endothelial cells forming microvessel walls, with the eventual goal of contributing to physiological understanding and medical progress.

DISCLOSURE STATEMENT

The author is not aware of any biases that might be perceived as affecting the objectivity of this review.

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